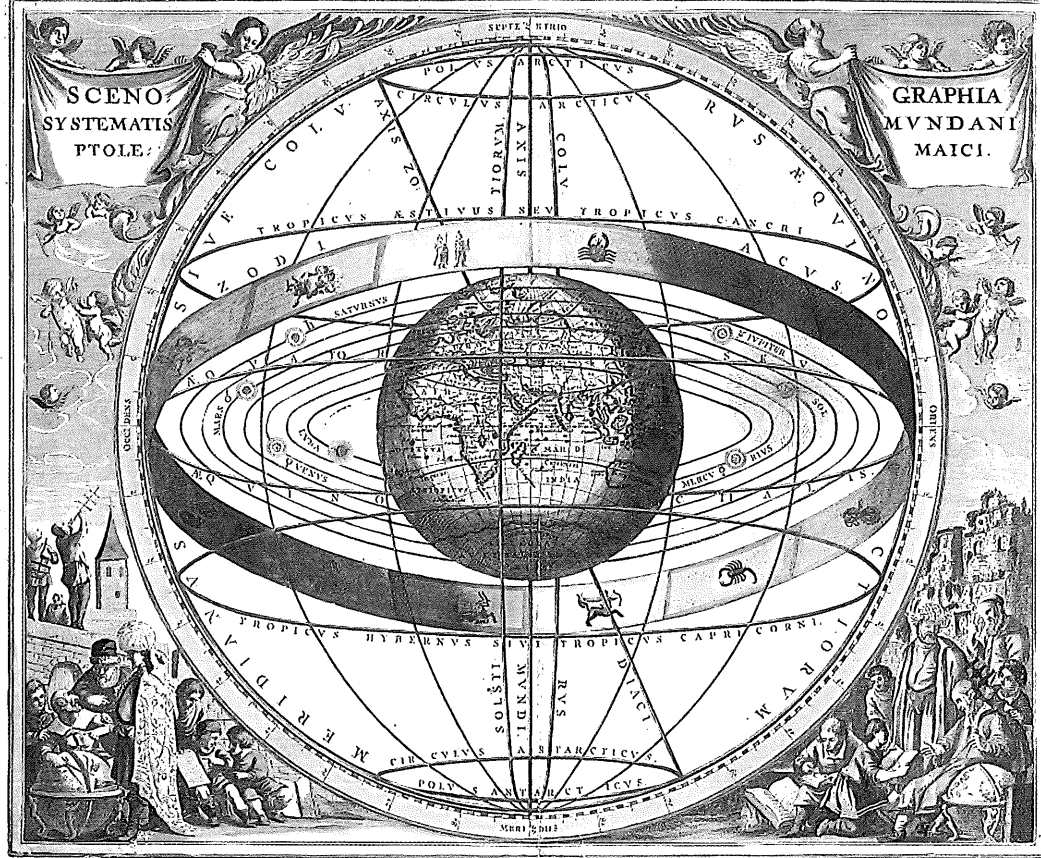


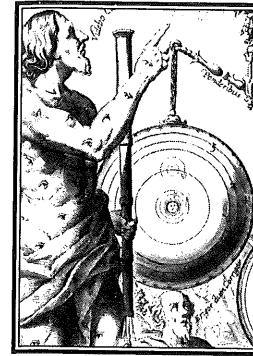


PHILOSOPHY OF SCIENCE

THE CENTRAL ISSUES

Second Edition

MARTIN CURD | J.A. COVER
CHRISTOPHER PINCOCK



1 | Science and Pseudoscience

INTRODUCTION

Parapsychology is defined by its practitioners as the study of extrasensory perception (ESP) and paranormal powers such as telekinesis. ESP includes such alleged psychic phenomena as telepathy, clairvoyance, and precognition. Shunned for decades by the scientific establishment, parapsychologists received official recognition in 1969 when the American Association for the Advancement of Science (the AAAS) admitted the Parapsychological Association as an affiliate member. Many scientists are unhappy with this decision, since they regard parapsychology as a pseudoscience. In 1979, the renowned physicist John A. Wheeler wrote a blistering letter to the president of the AAAS urging that the parapsychologists be expelled from the association. Wheeler wrote, "We have enough charlatanism in this country today without needing a scientific organization to prostitute itself to it. The AAAS has to make up its mind whether it is seeking popularity or whether it is strictly a scientific organization."¹

The debate about the nature of science—about its scope, methods, and aims—is as old as science itself. But this debate becomes especially heated when one group of practitioners accuses another group of practicing pseudoscience. Many individuals, groups, and theories have been accused of being pseudoscientific, including Freud and psychoanalysis, astrology, believers in the paranormal, Immanuel Velikovsky and Erich von Daniken (whose best-selling books *Worlds in Collision* and *Chariots of the Gods* excited the wrath of Carl Sagan and the scientific establishment), and, most recently, the self-styled advocates of creation-science. The proponents of astrology, the paranormal, psychoanalysis, and creation-science engage in research, write books, and publish articles, but their work is typically found in popular magazines and bookstores rather than refereed journals and

science libraries. They are seldom funded by the National Science Foundation or elected to the National Academy of Sciences. They are outside of the scientific establishment and are kept out by those who regard themselves as real scientists.

If our only concern were to label certain people “pseudoscientists,” we might simply check where their work is published and how their theories have been received by the scientific community. But we are concerned with the reasons certain doctrines are considered pseudoscientific; it is those reasons that interest philosophers of science.

Some philosophers have proposed necessary conditions for genuine science. That is, they have offered characteristics that any discipline or field of study must possess in order to qualify as genuine science. These characteristics are often called *demarcation criteria* because they can be used to differentiate science from its counterfeit: if a discipline fails to meet one of these conditions, then it is judged to be nonscientific.

Philosophers of science have often disagreed about demarcation criteria. In this chapter Karl Popper, Thomas Kuhn, Imre Lakatos, and Paul Thagard each defend a different set of necessary conditions for genuine science. Popper’s view, that a scientific theory must be open to refutation by making testable predictions, has been very influential, especially among working scientists. Kuhn, Lakatos, and Thagard all reject Popper’s claim that falsifiability is the hallmark of genuine science but disagree about what should replace it. All three address whether a theory or discipline’s claim to scientific legitimacy depends on historical considerations, such as how theories have developed over time.

The chapter ends with an exchange of views between Michael Ruse and Larry Laudan about the credentials of creation-science. Ruse, a prominent philosopher of biology, served as an expert witness in a trial concerning the constitutionality of an Arkansas law requiring public school biology teachers to present creationism as a viable scientific alternative to evolutionary theory. Under Ruse’s guidance, the judge in the case drew up a list of five criteria for genuine science and concluded that creation-science failed on all five counts. Laudan not only criticizes the items on this list (which includes Popper’s falsifiability) but also doubts whether there are any demarcation criteria that all scientific theories must satisfy.

■ | Note

1. Quoted in Jack W. Grove, *In Defence of Science* (Toronto: University of Toronto Press, 1989), 137. See also Martin Gardner, *Science: Good, Bad and Bogus* (Buffalo, N.Y.: Prometheus Books, 1981), 185–206, and Richard Wiseman, *Paranormality: Why We See What Isn’t There* (London: Macmillan, 2011). The Parapsychological Association is still a member of the AAAS.

KARL POPPER

Science: Conjectures and Refutations

Mr. Turnbull had predicted evil consequences, . . . and was now doing the best in his power to bring about the verification of his own prophecies.

—Anthony Trollope [*Phineas Finn*, ch. 25]

■ | I

When I received the list of participants in this course and realized that I had been asked to speak to philosophical colleagues* I thought, after some hesitation and consultation, that you would probably prefer me to speak about those problems which interest me most, and about those developments with which I am most intimately acquainted. I therefore decided to do what I have never done before: to give you a report on my own work in the philosophy of science, since the autumn of 1919 when I first began to grapple with the problem, ‘*When should a theory be ranked as scientific?*’ or ‘*Is there a criterion for the scientific character or status of a theory?*’

The problem which troubled me at the time was neither, ‘When is a theory true?’ nor, ‘When is a theory acceptable?’ My problem was different. I wished to distinguish between science and pseudo-science; knowing very well that science often errs, and that pseudo-science may happen to stumble on the truth.

I knew, of course, the most widely accepted answer to my problem: that science is distinguished from pseudo-science—or from ‘metaphysics’—by its *empirical method*, which is essentially *inductive*, proceeding from observation

FROM Karl Popper, *Conjectures and Refutations* (London: Routledge & Kegan Paul, 1963), 33–39.

* This essay was originally presented as a lecture at Peterhouse College at Cambridge University in the summer of 1953 as part of a course on developments and trends in contemporary British philosophy, organized by the British Council. It was originally published as “Philosophy of Science: A Personal Report,” in *British Philosophy in Mid-Century*, ed. C. A. Mace (London: Allen & Unwin, 1957).

or experiment. But this did not satisfy me. On the contrary, I often formulated my problem as one of distinguishing between a genuinely empirical method and a non-empirical or even a pseudo-empirical method—that is to say, a method which, although it appeals to observation and experiment, nevertheless does not come up to scientific standards. The latter method may be exemplified by astrology, with its stupendous mass of empirical evidence based on observation—on horoscopes and on biographies.

But as it was not the example of astrology which led me to my problem I should perhaps briefly describe the atmosphere in which my problem arose and the examples by which it was stimulated. After the collapse of the Austrian Empire there had been a revolution in Austria: the air was full of revolutionary slogans and ideas, and new and often wild theories. Among the theories which interested me Einstein's theory of relativity was no doubt by far the most important. Three others were Marx's theory of history, Freud's psycho-analysis, and Alfred Adler's so-called 'individual psychology'.*

There was a lot of popular nonsense talked about these theories, and especially about relativity (as still happens even today), but I was fortunate in those who introduced me to the study of this theory. We all—the small circle of students to which I belonged—were thrilled with the result of Eddington's eclipse observations which in 1919 brought the first important confirmation of Einstein's theory of gravitation. It was a great experience for us, and one which had a lasting influence on my intellectual development.†

The three other theories I have mentioned were also widely discussed among students at that time. I myself happened to come into personal contact with Alfred Adler, and even to co-operate with him in his social work among the children and young people in the working-class districts of Vienna where he had established social guidance clinics.

It was during the summer of 1919 that I began to feel more and more dissatisfied with these three theories—the Marxist theory of history, psycho-analysis, and individual psychology; and I began to feel dubious about their claims to scientific status. My problem perhaps first took the simple form, 'What is wrong with Marxism, psycho-analysis, and individual psychology?'

* For a fascinating autobiographical account of Popper's youthful flirtation and painful disenchantment with Marxism, see "A Crucial Year: Marxism; Science and Pseudoscience," in *The Philosophy of Karl Popper*, ed. Paul A. Schilpp (La Salle, Ill.: Open Court, 1974), 1:23–29. There is also an extended criticism of Freud in Karl R. Popper, *Realism and the Aim of Science* (New York: Routledge, 1983), 163–74.

† Einstein's general theory of relativity entails that light rays must bend in a gravitational field. Organized by Sir Arthur Eddington, two Royal Astronomical Society expeditions were dispatched to observe the solar eclipse of 1919, and verified that starlight was indeed deflected by the sun by the amount that Einstein had predicted. The *Times* of London reported this success as the most remarkable scientific event since the discovery of the planet Neptune. The light-bending test of relativity theory is discussed in "Popper's Demarcation Criterion," in the commentary on chapter 1, and in "Two Arguments for Explanationism," in the commentary on chapter 4.

Why are they so different from physical theories, from Newton's theory, and especially from the theory of relativity?'

To make this contrast clear I should explain that few of us at the time would have said that we believed in the *truth* of Einstein's theory of gravitation. This shows that it was not my doubting the *truth* of those other three theories which bothered me, but something else. Yet neither was it that I merely felt mathematical physics to be more *exact* than the sociological or psychological type of theory. Thus what worried me was neither the problem of truth, at that stage at least, nor the problem of exactness or measurability. It was rather that I felt that these other three theories, though posing as sciences, had in fact more in common with primitive myths than with science; that they resembled astrology rather than astronomy.

I found that those of my friends who were admirers of Marx, Freud, and Adler, were impressed by a number of points common to these theories, and especially by their apparent *explanatory power*. These theories appeared to be able to explain practically everything that happened within the fields to which they referred. The study of any of them seemed to have the effect of an intellectual conversion or revelation, opening your eyes to a new truth hidden from those not yet initiated. Once your eyes were thus opened you saw confirming instances everywhere: the world was full of *verifications* of the theory. Whatever happened always confirmed it. Thus its truth appeared manifest; and unbelievers were clearly people who did not want to see the manifest truth; who refused to see it, either because it was against their class interest, or because of their repressions which were still 'un-analysed' and crying aloud for treatment.

The most characteristic element in this situation seemed to me the incessant stream of confirmations, of observations which 'verified' the theories in question; and this point was constantly emphasized by their adherents. A Marxist could not open a newspaper without finding on every page confirming evidence for his interpretation of history; not only in the news, but also in its presentation—which revealed the class bias of the paper—and especially of course in what the paper did *not* say. The Freudian analysts emphasized that their theories were constantly verified by their 'clinical observations'. As for Adler, I was much impressed by a personal experience. Once, in 1919, I reported to him a case which to me did not seem particularly Adlerian, but which he found no difficulty in analysing in terms of his theory of inferiority feelings, although he had not even seen the child. Slightly shocked, I asked him how he could be so sure. 'Because of my thousandfold experience,' he replied; whereupon I could not help saying: 'And with this new case, I suppose, your experience has become thousand-and-one-fold.'

What I had in mind was that his previous observations may not have been much sounder than this new one; that each in its turn had been interpreted in the light of 'previous experience', and at the same time counted as additional confirmation. What, I asked myself, did it confirm? No more

than that a case could be interpreted in the light of the theory. But this meant very little, I reflected, since every conceivable case could be interpreted in the light of Adler's theory, or equally of Freud's. I may illustrate this by two very different examples of human behaviour: that of a man who pushes a child into the water with the intention of drowning it; and that of a man who sacrifices his life in an attempt to save the child. Each of these two cases can be explained with equal ease in Freudian and in Adlerian terms. According to Freud the first man suffered from repression (say, of some component of his Oedipus complex), while the second man had achieved sublimation. According to Adler the first man suffered from feelings of inferiority (producing perhaps the need to prove to himself that he dared to commit some crime), and so did the second man (whose need was to prove to himself that he dared to rescue the child). I could not think of any human behaviour which could not be interpreted in terms of either theory. It was precisely this fact—that they always fitted, that they were always confirmed—which in the eyes of their admirers constituted the strongest argument in favour of these theories. It began to dawn on me that this apparent strength was in fact their weakness.

With Einstein's theory the situation was strikingly different. Take one typical instance—Einstein's prediction, just then confirmed by the findings of Eddington's expedition. Einstein's gravitational theory had led to the result that light must be attracted by heavy bodies (such as the sun), precisely as material bodies were attracted. As a consequence it could be calculated that light from a distant fixed star whose apparent position was close to the sun would reach the earth from such a direction that the star would seem to be slightly shifted away from the sun; or, in other words, that stars close to the sun would look as if they had moved a little away from the sun, and from one another. This is a thing which cannot normally be observed since such stars are rendered invisible in daytime by the sun's overwhelming brightness; but during an eclipse it is possible to take photographs of them. If the same constellation is photographed at night one can measure the distances on the two photographs, and check the predicted effect.

Now the impressive thing about this case is the *risk* involved in a prediction of this kind. If observation shows that the predicted effect is definitely absent, then the theory is simply refuted. The theory is *incompatible with certain possible results of observation*—in fact with results which everybody before Einstein would have expected.¹ This is quite different from the situation I have previously described, when it turned out that the theories in question were compatible with the most divergent human behaviour, so that it was practically impossible to describe any human behaviour that might not be claimed to be a verification of these theories.

These considerations led me in the winter of 1919–20 to conclusions which I may now reformulate as follows.

- 1 It is easy to obtain confirmations, or verifications, for nearly every theory—if we look for confirmations.
- 2 Confirmations should count only if they are the result of *risky predictions*; that is to say, if, unenlightened by the theory in question, we should have expected an event which was incompatible with the theory—an event which would have refuted the theory.
- 3 Every 'good' scientific theory is a prohibition: it forbids certain things to happen. The more a theory forbids, the better it is.
- 4 A theory which is not refutable by any conceivable event is non-scientific. Irrefutability is not a virtue of a theory (as people often think) but a vice.
- 5 Every genuine *test* of a theory is an attempt to falsify it, or to refute it. Testability is falsifiability; but there are degrees of testability: some theories are more testable, more exposed to refutation, than others; they take, as it were, greater risks.
- 6 Confirming evidence should not count *except when it is the result of a genuine test of the theory*; and this means that it can be presented as a serious but unsuccessful attempt to falsify the theory. (I now speak in such cases of 'corroborating evidence'.)
- 7 Some genuinely testable theories, when found to be false, are still upheld by their admirers—for example by introducing *ad hoc* some auxiliary assumption, or by re-interpreting the theory *ad hoc* in such a way that it escapes refutation. Such a procedure is always possible, but it rescues the theory from refutation only at the price of destroying, or at least lowering, its scientific status. (I later described such a rescuing operation as a '*conventionalist twist*' or a '*conventionalist stratagem*'.)

One can sum up all this by saying that *the criterion of the scientific status of a theory is its falsifiability, or refutability, or testability*.

■ | II

I may perhaps exemplify this with the help of the various theories so far mentioned. Einstein's theory of gravitation clearly satisfied the criterion of falsifiability. Even if our measuring instruments at the time did not allow us to pronounce on the results of the tests with complete assurance, there was clearly a possibility of refuting the theory.

Astrology did not pass the test. Astrologers were greatly impressed, and misled, by what they believed to be confirming evidence—so much so that they were quite unimpressed by any unfavourable evidence. Moreover, by making their interpretations and prophecies sufficiently vague they were

able to explain away anything that might have been a refutation of the theory had the theory and the prophecies been more precise. In order to escape falsification they destroyed the testability of their theory. It is a typical soothsayer's trick to predict things so vaguely that the predictions can hardly fail: that they become irrefutable.

The Marxist theory of history, in spite of the serious efforts of some of its founders and followers, ultimately adopted this soothsaying practice. In some of its earlier formulations (for example in Marx's analysis of the character of the 'coming social revolution') their predictions were testable, and in fact falsified.² Yet instead of accepting the refutations the followers of Marx re-interpreted both the theory and the evidence in order to make them agree. In this way they rescued the theory from refutation; but they did so at the price of adopting a device which made it irrefutable. They thus gave a 'conventionalist twist' to the theory; and by this stratagem they destroyed its much advertised claim to scientific status.

The two psycho-analytic theories were in a different class. They were simply non-testable, irrefutable. There was no conceivable human behaviour which could contradict them. This does not mean that Freud and Adler were not seeing certain things correctly: I personally do not doubt that much of what they say is of considerable importance, and may well play its part one day in a psychological science which is testable. But it does mean that those 'clinical observations' which analysts naïvely believe confirm their theory cannot do this any more than the daily confirmations which astrologers find in their practice.³ And as for Freud's epic of the Ego, the Super-ego, and the Id, no substantially stronger claim to scientific status can be made for it than for Homer's collected stories from Olympus. These theories describe some facts, but in the manner of myths. They contain most interesting psychological suggestions, but not in a testable form.

At the same time I realized that such myths may be developed, and become testable; that historically speaking all—or very nearly all—scientific theories originate from myths, and that a myth may contain important anticipations of scientific theories. Examples are Empedocles' theory of evolution by trial and error, or Parmenides' myth of the unchanging block universe in which nothing ever happens and which, if we add another dimension, becomes Einstein's block universe (in which, too, nothing ever happens, since everything is, four-dimensionally speaking, determined and laid down from the beginning). I thus felt that if a theory is found to be non-scientific, or 'metaphysical' (as we might say), it is not thereby found to be unimportant, or insignificant, or 'meaningless', or 'nonsensical'.⁴ But it cannot claim to be backed by empirical evidence in the scientific sense—although it may easily be, in some genetic sense, the 'result of observation'.

(There were a great many other theories of this pre-scientific or pseudo-scientific character, some of them, unfortunately, as influential as the Marxist interpretation of history; for example, the racist interpretation of history—

another of those impressive and all-explanatory theories which act upon weak minds like revelations.)

Thus the problem which I tried to solve by proposing the criterion of falsifiability was neither a problem of meaningfulness or significance, nor a problem of truth or acceptability. It was the problem of drawing a line (as well as this can be done) between the statements, or systems of statements, of the empirical sciences, and all other statements—whether they are of a religious or of a metaphysical character, or simply pseudo-scientific. Years later—it must have been in 1928 or 1929—I called this first problem of mine the '*problem of demarcation*'. The criterion of falsifiability is a solution to this problem of demarcation, for it says that statements or systems of statements, in order to be ranked as scientific, must be capable of conflicting with possible, or conceivable, observations. . . .

■ | Notes

1. This is a slight oversimplification, for about half of the Einstein effect may be derived from the classical theory, provided we assume a ballistic theory of light.
2. See, for example, my *Open Society and Its Enemies* [Routledge & Kegan Paul, 1945], ch. 15, section iii, and notes 13–14.
3. 'Clinical observations', like all other observations, are *interpretations in the light of theories* . . . ; and for this reason alone they are apt to seem to support those theories in the light of which they were interpreted. But real support can be obtained only from observations undertaken as tests (by 'attempted refutations'); and for this purpose *criteria of refutation* have to be laid down beforehand: it must be agreed which observable situations, if actually observed, mean that the theory is refuted. But what kind of clinical responses would refute to the satisfaction of the analyst not merely a particular analytic diagnosis but psycho-analysis itself? And have such criteria ever been discussed or agreed upon by analysts? Is there not, on the contrary, a whole family of analytic concepts, such as 'ambivalence' (I do not suggest that there is no such thing as ambivalence), which would make it difficult, if not impossible, to agree upon such criteria? Moreover, how much headway has been made in investigating the question of the extent to which the (conscious or unconscious) expectations and theories held by the analyst influence the 'clinical responses' of the patient? (To say nothing about the conscious attempts to influence the patient by proposing interpretations to him, etc.) Years ago I introduced the term '*Oedipus effect*' to describe the influence of a theory or expectation or prediction upon the event which it predicts or describes: it will be remembered that the causal chain leading to Oedipus' parricide was started by the oracle's prediction of this event. This is a characteristic and recurrent theme of such myths, but one which seems to have failed to attract the interest of the analysts, perhaps not accidentally. (The problem of confirmatory dreams suggested by the analyst is discussed by Freud, for example in *Gesammelte Schriften* [Complete works], III, 1925, where he says on p. 314: 'If anybody asserts that most of the dreams which can be utilized in an analysis . . . owe their origin to [the analyst's] suggestion, then no objection can be made from the

point of view of analytic theory. Yet there is nothing in this fact', he surprisingly adds, 'which would detract from the reliability of our results.')

4. The case of astrology, nowadays a typical pseudo-science, may illustrate this point. It was attacked, by Aristotelians and other rationalists, down to Newton's day, for the wrong reason—for its now accepted assertion that the planets had an 'influence' upon terrestrial ('sublunar') events. In fact Newton's theory of gravity, and especially the lunar theory of the tides, was historically speaking an offspring of astrological lore. Newton, it seems, was most reluctant to adopt a theory which came from the same stable as for example the theory that 'influenza' epidemics are due to an astral 'influence'. And Galileo, no doubt for the same reason, actually rejected the lunar theory of the tides; and his misgivings about Kepler may easily be explained by his misgivings about astrology.